

Round trips are not enough

Efficiency of non-reversible parallel tempering on inhomogeneous multimodal targets

Han Myo Htet · supervised by Professor Krzysztof Łatuszyński · Department of Statistics, University of Warwick · URSS 2026

1 Sampling many modes

Posteriors in Bayesian inference and the loss surfaces of neural networks are routinely **multimodal**. A single Metropolis chain settles into one mode and then essentially never leaves it. Our benchmark makes the difficulty explicit and controllable:

$$\pi(x) = \frac{1}{2}\mathcal{N}(\mathbf{1}_d, s_1^2 I) + \frac{1}{2}\mathcal{N}(-\mathbf{1}_d, s_2^2 I), \quad x \in \mathbb{R}^d.$$

Both modes carry weight exactly $\frac{1}{2}$. When $s_1 = s_2$ the target is *homogeneous*; when $s_1 \neq s_2$ the two modes have different shapes, and that single difference is what the whole poster is about.

Parallel tempering runs N chains along a temperature ladder $1 = \beta_0 > \beta_1 > \dots > \beta_{N-1} = \beta_{\min}$, chain j targeting

$$\pi_{\beta_j}(x) \propto \pi(x)^{\beta_j}.$$

Flattened chains cross barriers easily; swap moves pass that mobility down to the cold chain $\beta_0 = 1$, which is the one we keep.

2 The two schedules

Algorithm 1 One parallel tempering scan

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1 input: ladder  $\beta$ , states  $x_0, \dots, x_{N-1}$ , scan index  $i$ 
2 for  $j = 0, \dots, N-1$  in parallel
3   for  $t = 1, \dots, T$ 
4      $y \sim \mathcal{N}(x_j, (r^2/\beta_j)I)$ 
5      $x_j \leftarrow y$  w.p.  $1 \wedge (\pi(y)/\pi(x_j))^{\beta_j}$ 
6 SEO :  $p \sim \text{Bernoulli}(1/2)$ 
7 DEO :  $p \leftarrow i \bmod 2$ 
8 for  $j = p, p+2, \dots \leq N-2$ 
9   swap  $x_j \leftrightarrow x_{j+1}$  w.p.  $1 \wedge A_j$ 
10 record the chain each replica now occupies
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The two schemes differ only in **lines 6–7**. The swap acceptance ratio is

$$A_j = \frac{\pi(x_j)^{\beta_{j+1}} \pi(x_{j+1})^{\beta_j}}{\pi(x_j)^{\beta_j} \pi(x_{j+1})^{\beta_{j+1}}} = \exp\{(\beta_{j+1} - \beta_j)[\log \pi(x_j) - \log \pi(x_{j+1})]\}.$$

Under **SEO** the parity is random, so a replica's position on the ladder performs a *random walk* and needs order N^2 scans to cross it. Under **DEO** the parity alternates deterministically, replicas move *ballistically*, and they cross in order N scans [1].

A NOTE ON IMPLEMENTATION

Forming A_j or the local acceptance ratio from densities rather than *log* densities underflows at $d = 30$: below $\beta \approx 0.02$ both $\pi(x)$ and $\pi(y)$ evaluate to exactly zero, the ratio becomes 0/0, and the hottest $\sim 15\%$ of the ladder silently accepts every proposal. All results here use log densities.

3 What the theory promises

Write s_j for the acceptance rate of the swap between chains j and $j+1$, and $r_j = 1 - s_j$ for the rejection rate. Syed et al. [1] show efficiency is governed by the **communication barrier**

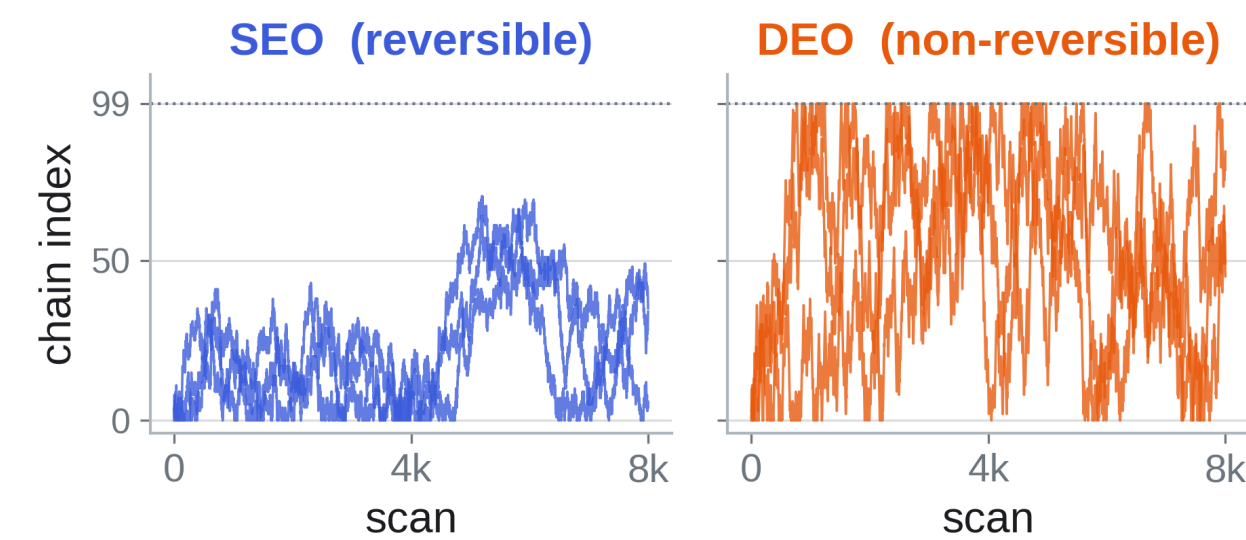
$$\Lambda = \sum_{j=0}^{N-2} r_j \xrightarrow{N \rightarrow \infty} \int_{\beta_{\min}}^1 \lambda(\beta) d\beta,$$

a property of the temperature *path*, not of the schedule. With the schedule inefficiency $E = \sum_j r_j/s_j$, the **round trip rate** τ — complete target $\rightarrow$ reference $\rightarrow$ target excursions per scan — obeys

$$\tau_{\text{SEO}} = \frac{1}{2N + 2E}, \quad \tau_{\text{DEO}} = \frac{1}{2 + 2E},$$

$$\tau_{\text{DEO}} \xrightarrow{N \rightarrow \infty} \bar{\tau} = \frac{1}{2 + 2\Lambda}.$$

So refining the ladder drives τ_{SEO} to zero like $1/N$, while τ_{DEO} approaches a limit **independent of N** , maximised at $N \approx 2\Lambda$. That is the promise of non-reversibility, and the thing we set out to test.

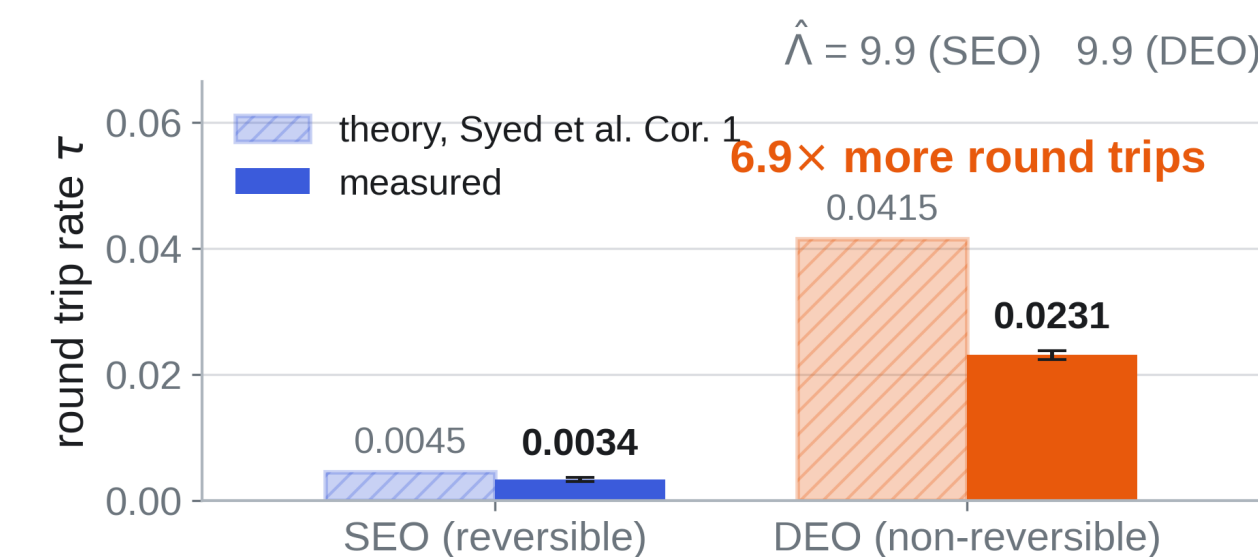


Chain index 0 is the target, 99 the reference. Under DEO a replica sweeps the whole ladder; under SEO it diffuses near the cold end and rarely reaches the top.

THE EXPERIMENT

$d = 30$ · $s_1 = 0.3$ (mode A, at +1) · $s_2 = 0.4$ (mode B, at -1) · $N = 100$ chains · β geometric on $[0.01, 1]$ · $r = 0.15$, $T = 8$ · every chain started in mode B · densities evaluated in **log space**.

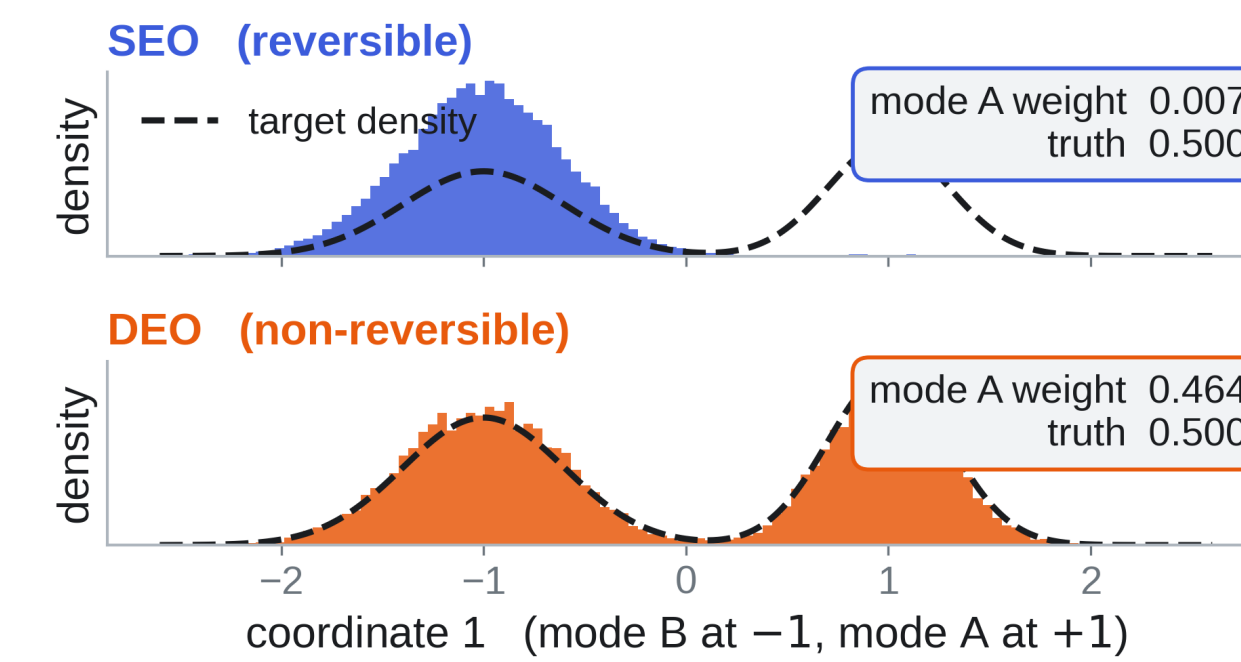
4 The speed-up is real



Both schedules recover the **same barrier**, as they must. DEO completes **7x more round trips**. Both rates sit below the Corollary 1 values, which an optimally tuned ladder attains at stationarity; the qualitative prediction holds clearly.

5 But the samples are wrong

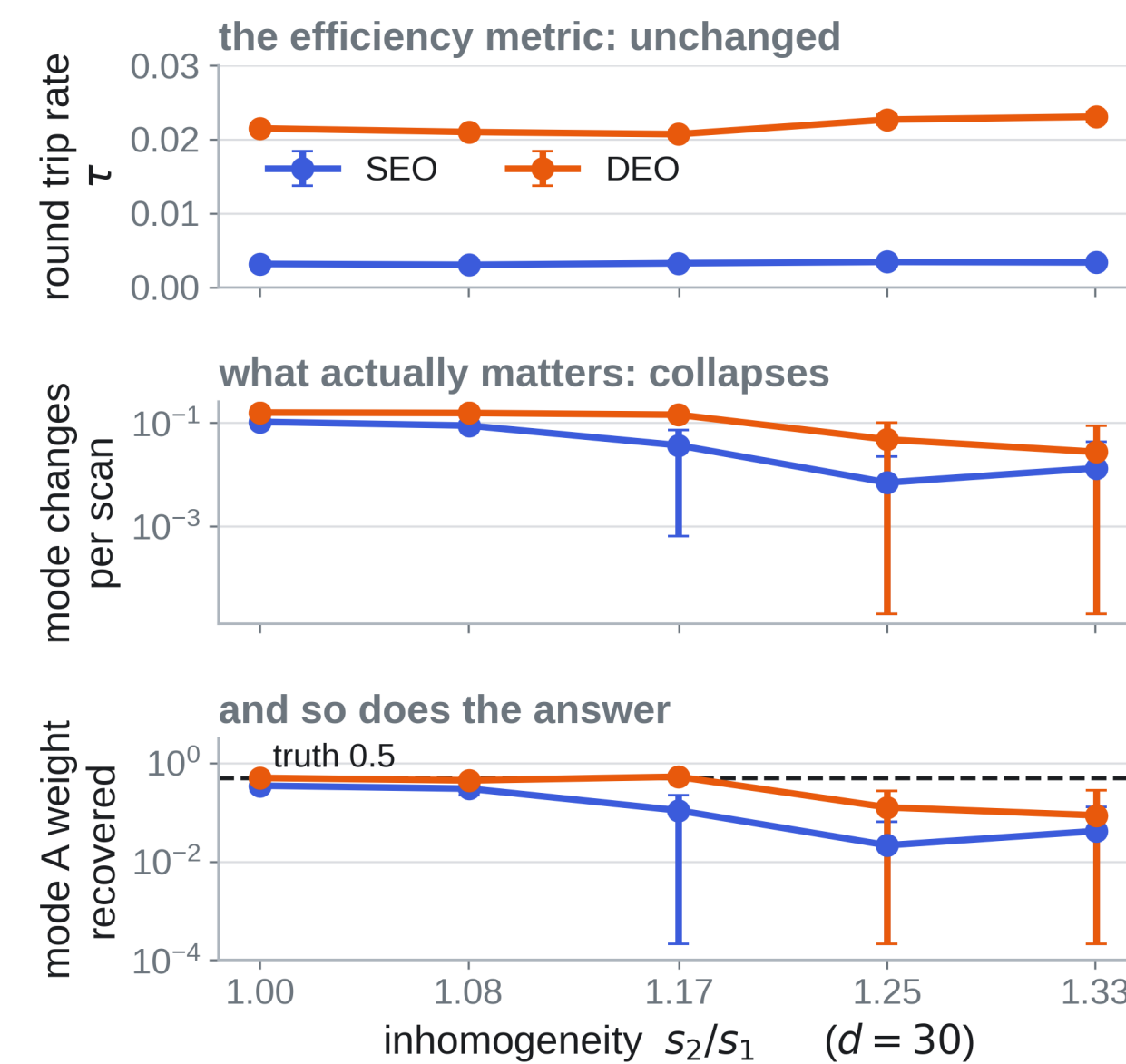
$d = 30$, $s_1 = 0.3$, $s_2 = 0.4$, 25 000 scans



Same target, same budget, same ladder. SEO never escapes the mode it started in; DEO does. Yet the round trip rate separates the two by a factor of 7, while the recovered mode weight separates them by a factor of 65. **τ does not predict the gap it is meant to explain.**

6 A controlled test

Hold the algorithm, the ladder and the budget fixed, and turn one knob: the **inhomogeneity** s_2/s_1 , from 1 (identical modes) to 4/3. Both modes keep weight $\frac{1}{2}$ throughout — the right answer never moves.



Mean over 6 independent runs of 20 000 scans; bars are ± 1 s.d.

The round trip rate is **flat to within $\pm 6\%$** across the sweep, for both schedules, while the recovered weight of mode A falls from 0.35 to 0.042 under SEO. Judged by its own diagnostic the sampler is in identical health at both ends. **The metric cannot see the difficulty it is meant to measure.**

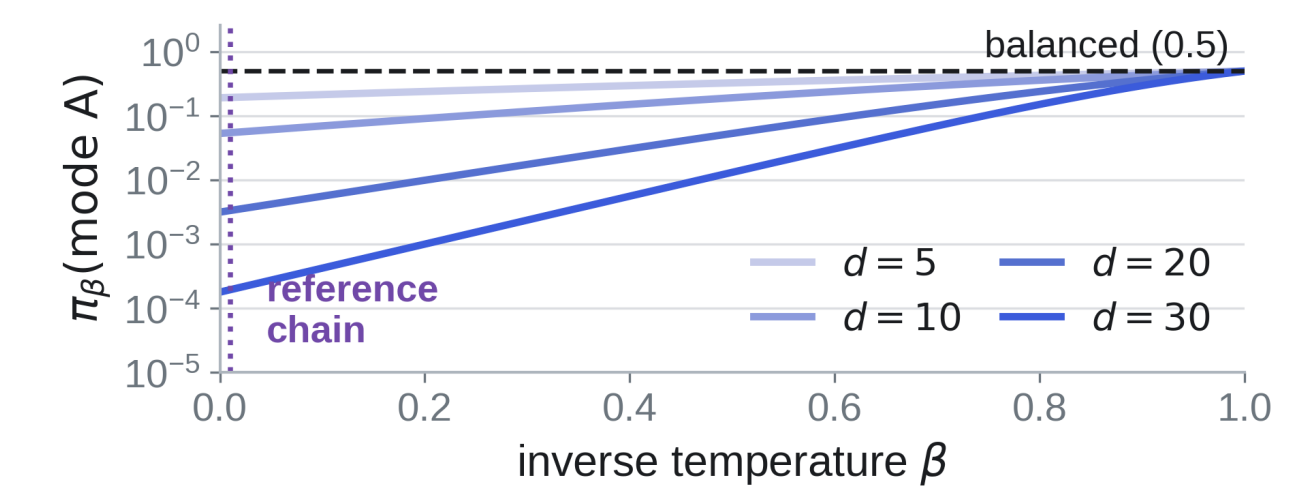
7 Why the metric is blind

Take the modes well separated and apply Laplace's approximation to π^β . Near mode i ,

$$\pi(x)^\beta \approx \left[\frac{1}{2} (2\pi s_i^2)^{-d/2} \right]^\beta \exp\left\{ -\frac{\beta}{2s_i^2} \|x - \nu_i\|^2 \right\},$$

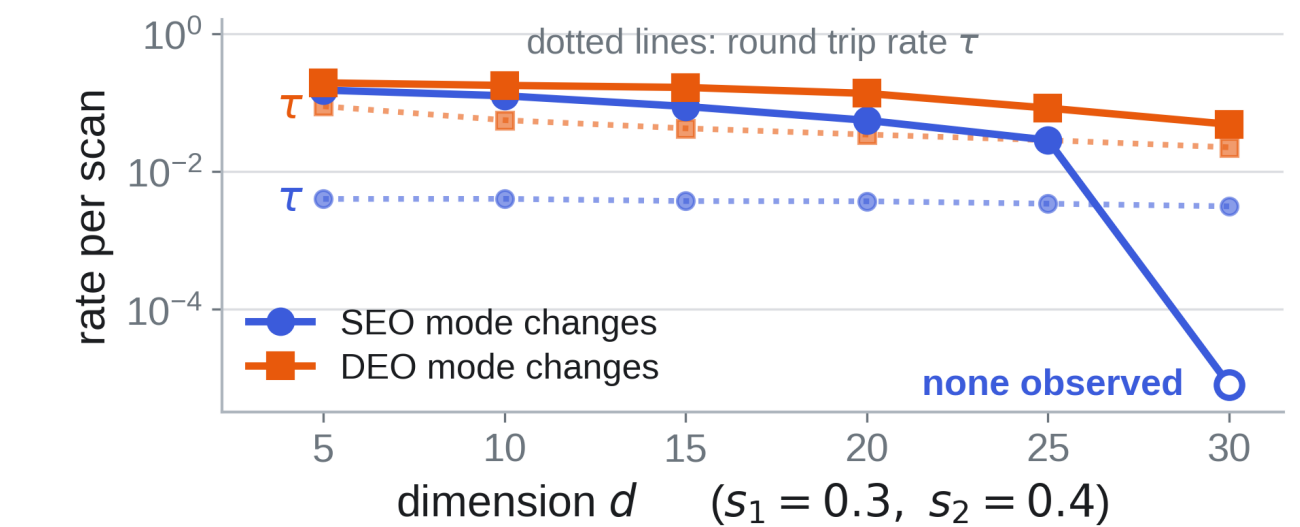
a Gaussian of variance s_i^2/β . Integrating gives mass proportional to $(s_i^2)^{-d\beta/2} (s_i^2/\beta)^{d/2}$; the β factors cancel in the ratio and

$$\frac{\pi_\beta(A)}{\pi_\beta(B)} = \left(\frac{s_1}{s_2} \right)^{d(1-\beta)}$$



Only at $\beta = 1$ are the modes balanced, and the imbalance is **exponential in $d(1-\beta)$** : at $d = 30$ the reference chain gives mode A a weight of 1.9×10^{-4} instead of $\frac{1}{2}$. A replica that climbs the whole ladder is therefore very likely to come back down into the mode it left: round trips are plentiful, **useful** round trips are not. This is the mechanism behind the torpid-mixing bounds of Woodard et al. [2].

8 Blindness in dimension



Fix $s_2/s_1 = 4/3$ and raise d instead. The SEO round trip rate falls by only $1.3\times$, while its mode-change rate falls to zero: in four runs of 20 000 scans at $d = 30$ the cold chain never once changed basin.

9 Conclusions

The non-reversible speed-up is real — an order-of-magnitude gain in round trips, as advertised.

τ and Λ **describe the ladder, not the target**, staying flat where the sampler goes from working to failing outright.

Refining the ladder cannot repair this — the imbalance belongs to the path, not its discretisation.

Report a mode-aware diagnostic alongside τ — how often the cold chain actually changes basin.